E1.07 – Power Rangers

Product Specification

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Table of Contents

1 Abstract 4

2 Keywords 4

3 Introduction 5

3.1 Motivation & Problem Statement 5

3.2 Scope of Work 5

3.3 Design Objective 5

4 Theoretical Background & Standards 7

4.1 Operating Principles 7

4.2 Electrical Standards 7

5 System Architecture 8

5.1 Top-Level Block Diagram 8

5.2 Interface Definition 8

6 Functional Description 9

6.1 Solar and wind Energy Subsystem 9

6.2 Solar and Wind Energy Design Analysis 9

6.3 Solar and Wind Energy Block Diagram (or Flow Diagram if applicable) 10

6.3.1 Solar and Wind Energy Interface 11

6.4 GaN-Based Boost Converter with CC/CV Battery Charging Subsystem 11

6.4.1 GaN-Based Boost Converter with CC/CV Battery Charging Design Analysis 11

6.4.2 GaN-Based Boost Converter with CC/CV Battery Charging Block Diagram 13

6.4.3 GaN-Based Boost Converter with CC/CV Battery Charging Interface 13

6.5 Real-Time Monitoring and Display Subsystem 14

6.5.1 Real-Time Monitoring and Display Design Analysis 14

6.5.2 Real-Time Monitoring and Display Block Diagram 15

6.5.3 Real-Time Monitoring and Display Interface 16

6.6 GaN-Based Buck Converter and 5 V Power Regulation Subsystem 16

6.6.1 GaN-Based Buck Converter and 5 V Power Regulation Design Analysis 16

6.6.2 GaN-Based Buck Converter and 5 V Power Regulation Block Diagram 18

6.6.3 GaN-Based Buck Converter and 5 V Power Regulation Interface 18

6.7 GaN device fabrication Subsystem 18

6.7.1 GaN device fabrication Design Analysis 19

6.7.2 GaN device fabrication Block Diagram 20

6.7.3 GaN device fabrication Interface 22

6.8 <Insert a meaningful name here> Subsystem 23
6.8.1 <Subsystem Name> Design Analysis 24
6.8.2 <Subsystem Name> Block Diagram (or Flow Diagram if applicable) 24
6.8.3 <Subsystem Name> Interface 24
6.9 Safety, Security and Administrative Functions 22
7 Conclusion 24
8 Project Alignment Matrix 24
9 References 26
10 Approvals 27
10.1 Contingencies 27
11 Appendix A: Test Procedures 28
11.1 Solar and Wind Energy Harvesting Subsystem Unit Tests 30
11.1.1 Solar Panel Output Characterization 30
11.1.2 Wind Generator Output Test 30
11.2 GaN-Based Boost Converter and CC/CV Battery Charging Subsystem Unit Tests 32
11.2.1 GaN-Based Boost Converter and CC/CV Battery Charging 32
11.3 Real-Time Monitoring and Display Subsystem Unit Tests 34
11.3.1 100 kHz PWM Frequency and Duty-Cycle Verification 34
11.3.2 Hard Safety Limit Trip Test 34
11.3.3 INA226 Sensor Accuracy Verification 35
11.3.4 Touch Tab Navigation and Relay Control Test 35
11.4 GaN-Based Buck Converter and 5 V Power Regulation Subsystem Unit Tests 37
11.4.1 GaN-Based GaN-Based Buck Converter and 5 V Power Regulation 37

1 Abstract

The Abstract was written by Alok Parajuli.

Our product is a hybrid solar-wind power conversion and energy storage system that combines renewable DC power sources with GaN-based power converters to charge a battery and supply regulated power to demonstration loads. The system uses a 12 W solar array and a mechanically driven wind generation subsystem as its input sources. A GaN-based boost converter with constant-current/constant-voltage (CC/CV) control is designed to charge a 12.8 V, 5 Ah lithium iron phosphate (LiFePO4) battery, while a GaN-based buck converter regulates the battery voltage to a 5 V DC bus.

A microcontroller-based monitoring system will measure voltage and current and display operating information. The system will be integrated into a physical smart-city model representing residential buildings, an electric vehicle charging station, and a data center. Testing will evaluate renewable-source output, converter operation, battery charging, monitoring accuracy, and 5 V output regulation. The project aims to demonstrate the practical application of GaN power semiconductor technology in a small-scale renewable-energy conversion and storage system.

2 Keywords

- Hybrid Renewable Energy System: A system that combines solar and wind energy sources to provide electrical power.

- Gallium Nitride (GaN): A wide-bandgap semiconductor material used in the project's power converters to control and convert electrical power.

- DC-DC Converter: An electronic circuit that converts one DC voltage level to another. This project uses boost and buck converters.

- Boost Converter: A DC-DC converter designed to increase the input voltage to a higher output voltage for battery charging.

- Buck Converter: A DC-DC converter designed to reduce the battery voltage to a regulated 5 V output for the demonstration loads.

- Constant-Current/Constant-Voltage (CC/CV) Charging: A charging method that regulates current during the initial charging stage and maintains a target voltage as the battery approaches full charge.

- Lithium Iron Phosphate (LiFePO4): A rechargeable battery chemistry selected for the system's energy storage, with a nominal voltage of 12.8 V and a capacity of 5 Ah.

- Pulse-Width Modulation (PWM): A switching control technique that adjusts the duty cycle of a signal to regulate power converter operation.

- Voltage and Current Monitoring: Measurement of electrical voltage and current to observe the performance of the renewable sources, converters, and battery.

- DC Power Distribution: The delivery of regulated DC power from the system's output bus to multiple demonstration loads.

3 Introduction

The Introduction was written by Alok Parajuli.

This project develops a hybrid solar-wind power conversion and energy storage system using GaN-based converters. It combines renewable DC sources, battery charging, regulated power delivery, and real-time monitoring in a physical smart-city model.

3.1 Motivation & Problem Statement

Solar and wind generation vary with environmental and operating conditions, creating challenges for battery charging and supplying stable power to electrical loads. This project investigates how power conversion, energy storage, and monitoring can address these challenges while demonstrating GaN-based power electronics. The resulting prototype will provide Texas State University with an educational demonstration of renewable-energy integration and power conversion.

3.2 Scope of Work

The project includes a nominal 12 W solar array, four R280 DC generator motors for mechanically driven wind generation, a GaN-based boost converter, a 12.8 V, 5 Ah LiFePO4 battery, a GaN-based buck converter, and microcontroller-based monitoring and display functions. The team will characterize the sources, develop and test the converter stages, implement charging control, and integrate the subsystems with representative smart-city loads. Commercial-scale generation, grid-connected operation, and functional high-power city loads are outside the project scope.

Table 1 summarizes the preliminary division of subsystem responsibilities among the team members. System integration and testing are shared responsibilities.

Team Member	Subsystem
Avash Aryal	GaN-Based Boost Converter; Gate Driver and CC/CV Battery Charging Control
Alok Parajuli	Solar and Wind Energy Harvesting; Source Characterization and Array Configuration
Ashok Paudel	Sensing, Monitoring and Control; Microcontroller, Electrical Sensors and Touchscreen Display
Michelle Su	Power Budget, LiFePO4 Battery Integration and Buck Converter

Table 1 Preliminary Subsystem Responsibilities

3.3 Design Objective

Table 2 Semester Deliverables

To be completed by 1st Semester Senior Design Day	☐ Complete solar and wind source characterization. ☐ Complete the boost converter PCB schematic and layout. ☐ Assemble the boost converter power stage and complete an initial open-loop bench test. ☐ Assemble the buck converter and verify its 5 V output.

	☐ Verify the Arduino CC/CV control loop and confirm closed-loop battery charging. ☐ Complete the wind generation stage mechanical assembly and characterization. ☐ Integrate the INA226 sensor array and touchscreen display. ☐ Integrate the solar source, wind source, boost converter, battery, buck converter, and display into the complete system. ☐ Complete subsystem-level bench testing, debugging, and initial full-system performance testing and validation.
To be completed by 2nd Semester Senior Design Day	☐ Complete GaN MOSFET fabrication for the boost and buck converter stages. ☐ Complete the enclosure and smart-city model load-network build-out. ☐ Demonstrate the full system's functionality. ☐ Complete formal load testing and efficiency evaluation across the planned operating modes. ☐ Complete PCB-level performance testing and validation. ☐ Present the final integrated prototype and document the final test results.

4 Theoretical Background & Standards

This section describes the electrical principles behind the hybrid solar-wind system, its power conversion stages, battery charging, and electrical monitoring. It also identifies relevant design and safety considerations for developing and testing the prototype.

4.1 Operating Principles

The solar panels convert incident light into DC electrical power through the photovoltaic effect. The four R280 DC motors act as generators when mechanically driven by the external motor, converting rotational mechanical energy into electrical energy. The output of both sources varies with operating conditions.

The boost converter uses a switching circuit and inductor to increase the DC input voltage. Its CC/CV charging control regulates the charging current and battery voltage for the 12.8 V LiFePO4 battery. The battery stores energy and supplies the input to the buck converter, which reduces the battery voltage to a regulated 5 V DC output.

The boost and buck converters use GaN power transistors as high-speed electronic switches. PWM controls the switching duty cycle to regulate converter operation. Voltage and current sensors measure electrical conditions, and the microcontroller processes the measurements for display and system monitoring.

4.2 Electrical Standards

Table 3: Appropriate Engineering Standard

Engineering Standard	Application of the Standard
EC 61010-1 -Electrical Measurement and Laboratory Equipment Safety	Provides a reference for safe electrical measurement and laboratory testing practices.
Battery Manufacturer Charging Specifications	Defines the allowable charging voltage and current for the 12.8 V LiFePO4 battery.
GaN Device and Component Datasheets	Establishes operating limits for the GaN transistors, gate drivers, inductors, diodes, and capacitors used in the converters.
I^2C and SPI Interface Specifications	Guides communication between the microcontroller, sensors, and touchscreen, where applicable.
Electrical Wiring and Overcurrent Protection Requirements	Guides wire sizing, fuse selection, and protection against excessive current and short circuits.

5 System Architecture

5.1 Top-Level Block Diagram

The top-level diagram of our system is shown in Figure 1. The system is divided into five main subsystems: wind generation, solar generation, power management and energy storage, monitoring and control, and load distribution. The solar and wind subsystems provide renewable energy input to the CC/CV boost converter, which regulates the power used to charge the LiFePO4 battery. The stored energy is then passed through a buck converter to provide a regulated 5 V DC bus for the model loads. The monitoring and control subsystem measures voltage and current at different points in the system and uses the Arduino to control and monitor the converters and overall system operation.

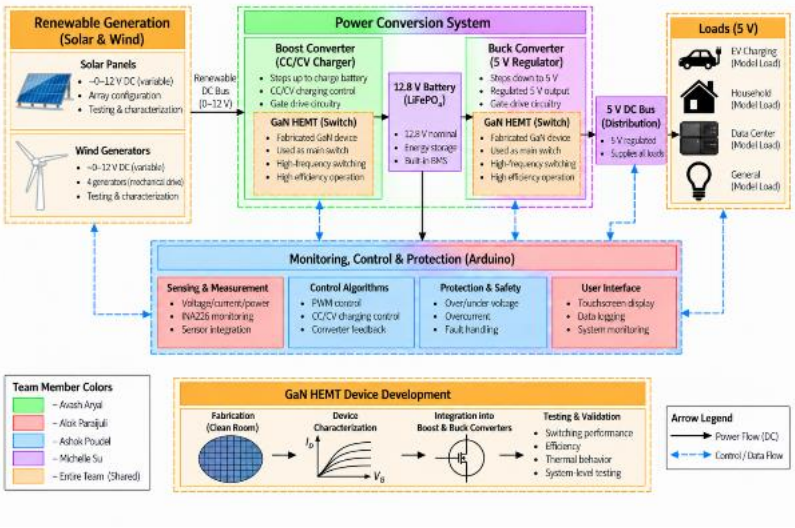


Figure 1: Top-Level Block Diagram

Key inputs to the system are the DC power generated by the solar and wind generation subsystems. The solar array is configured to provide approximately 12 V and up to 1 A under rated conditions, while the four R280 wind generators are designed to provide approximately 2–5 W of combined electrical power. These two renewable power sources are combined and supplied to the power conversion and energy storage subsystem. Key outputs of the system are a regulated 5 V DC supply for the model loads and real-time voltage, current, and power data displayed through the system interface.

5.2 Interface Definition

The system's five main subsystems exchange energy and control or monitoring signals through the connections shown in Figure 1. The solar panels and wind generators provide variable DC power to the renewable DC bus through separate isolation diodes. The bus supplies the GaN-based boost converter, which uses CC/CV control to charge the 12.8 V LiFePO4 battery. The battery then supplies the GaN-based buck converter, which regulates the output to the 5 V DC bus for the EV charging, household, data center, and general demonstration loads.

The Monitoring, Control & Protection (Arduino) subsystem connects to the other blocks through sensor measurements, converter-control signals, and relay commands. INA226 sensors provide voltage and current measurements for monitoring and feedback. The Arduino sends PWM signals through gate drivers to control the boost and buck converters, and uses digital outputs to operate the relays for the solar source, wind source, and demonstration loads. It also exchanges display data

and touch input with the touchscreen. The external motor mechanically drives the wind generators using a separate power supply and is not part of the renewable DC power path.

Interface	Connected blocks	Purpose
Renewable DC power	Solar and wind generation → isolation diodes → renewable DC bus	Combines the variable DC outputs of both sources while limiting reverse current between branches.
Charging power	Renewable DC bus → boost converter → 12.8 V battery	Converts the renewable input and provides CC/CV-regulated battery charging.
Load power	Battery → buck converter → 5 V DC bus → loads	Steps down the battery voltage and distributes regulated 5 V power to the demonstration loads.
Measurement and feedback	Renewable sources, DC bus, and battery → INA226 sensors → Arduino	Provides electrical measurements for monitoring and converter control.
PWM control	Arduino → gate drivers → boost and buck converters	Controls the switching operation of both converter stages.
Relay control	Arduino → relay module → solar, wind, and demonstration-load circuits	Enables the controller to switch the source and load circuits.
Display and touch	Arduino ↔ touchscreen	Displays system measurements and status and receives user input.

The system includes a controller logic domain, a nominal 12.8 V battery domain, and a regulated 5 V load domain. Control signals reach the converter switching devices through dedicated gate-driver circuits, while sensor and display interfaces connect to the controller at compatible logic levels. The external wind-drive motor has its own power supply, separate from the system's LiFePO4 energy-storage battery.

6 Functional Description

6.1 Solar and wind Energy Subsystem

The Solar and Wind Energy Subsystem provides the renewable DC power inputs for the hybrid energy system, using a 2S2P solar array and four belt-driven wind generators whose outputs are isolated and combined onto the common DC bus feeding the boost converter.

6.1.1 Solar and Wind Energy Design Analysis

The design analysis evaluates solar panel configuration, wind generator drive method, and source isolation based on confirmed array and generator output, reverse-current prevention, mechanical reliability, and compatibility with the common DC bus.

6.1.1.1 Solar and wind energy Objective

This subsystem provides the renewable DC power inputs for the hybrid energy system. The solar branch uses four 6 V, 3 W photovoltaic panels arranged in a 2-series, 2-parallel configuration, giving a nominal array rating of approximately 12 V and 12 W under rated conditions. The wind branch uses four R280 DC motors as generators, mechanically driven by an external motor through a belt-and-pulley arrangement. The output of each source will be characterized and combined through

isolation diodes before being supplied to the common DC bus for power conversion and battery charging.

6.1.1.2 Solar and Wind Energy Evaluation Criteria

Criterion	Why it matters
Solar output	Confirms the array produces usable voltage and current under available lighting conditions.
Wind generator output	Determines the actual voltage, current, and power produced by the mechanically driven generators.
Source isolation	Helps prevent unwanted reverse current between the solar and wind branches.
Mechanical reliability	The belt-and-pulley drive must rotate the generator motors consistently during testing.
Common DC bus compatibility	Both source branches must provide input power within the operating limits of the downstream converter.

Table 4: Evaluation Criteria

6.1.1.3 Solar and Wind Energy Design Alternatives

1. **Solar panel configuration:** Connect the four panels in a 2S2P arrangement or use a different series-parallel arrangement to meet the desired input voltage and current.
2. **Wind generator drive:** Use an external motor with a belt-and-pulley system or investigate alternative mechanical drive arrangements.
3. **Source isolation:** Use Schottky diodes to isolate the two source branches or use a more complex active isolation circuit.

6.1.1.4 Solar and Wind Energy Design Comparative Analysis and Conclusion

The 2S2P solar arrangement was selected to provide a nominal 12 V, 12 W source using the four available 6 V, 3 W panels. Actual output will depend on illumination and operating conditions.

For the wind subsystem, four R280 motors will be mechanically driven by an external motor through belts and pulleys. This approach provides a controllable way to demonstrate wind generation indoors without depending on natural wind. The electrical output of the generators will be measured during characterization; the final usable wind power has not yet been established.

Schottky diodes were selected for source isolation because they provide a simple way to reduce reverse current between the two source branches. The combined source output will feed the common DC bus and the downstream boost converter. The subsystem will be evaluated through voltage and current measurements and by checking the consistency of its output during operation.

6.1.2 Solar and Wind Energy Block Diagram (or Flow Diagram if applicable)

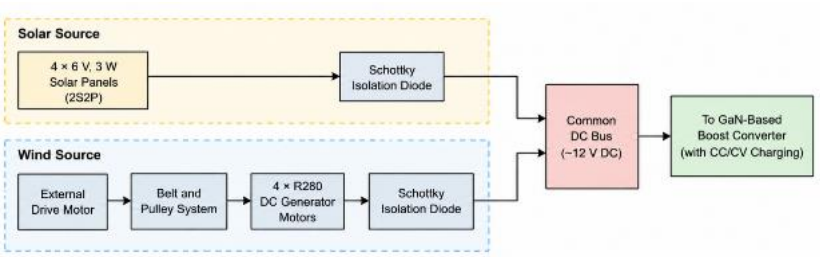


Figure 2: Solar and Wind Energy Harvesting Block Diagram

6.1.3 Solar and Wind Energy Interface

Inputs

Input	Source	Description
Solar energy	Available light	Energy converted into DC electrical output by the photovoltaic panels.
Mechanical rotation	External drive motor and belt-and-pulley mechanism	Rotates the R280 motors to generate electrical output.

Table 5: Inputs

Outputs

Output	Destination	Description
Solar DC output	Solar isolation diode and common DC bus	Variable DC power from the photovoltaic array.
Wind DC output	Wind isolation diode and common DC bus	Variable DC power from the generator motors.
Combined DC output	Boost converter input	Combined source power supplied for downstream conversion and battery charging.

Table 6: Outputs

6.2 GaN-Based Boost Converter with CC/CV Battery Charging Subsystem

The GaN-Based Boost Converter with CC/CV Battery Charging Subsystem converts variable solar and wind DC power into a regulated CC/CV charging profile for the 12.8 V LiFePO4 battery, using a GaN power stage with firmware-level control and an independent hard-limit safety check.

6.2.1 GaN-Based Boost Converter with CC/CV Battery Charging Design Analysis

The design analysis evaluates control algorithm, switching device, and boost diode selection based on control loop stability, fault coverage, component ratings, and code maintainability.

6.2.1.1 GaN-Based Boost Converter with CC/CV Battery Charging Objective

This subsystem regulates the variable DC power arriving from the solar and wind sources on the common DC bus, boosts it, and charges a 12.8 V LiFePO4 battery using a constant-current/constant-voltage (CC/CV) profile: 1.0 A constant current until the battery reaches 14.4 V, then holding 14.4 V while current tapers. The power stage (GaN switching device, inductor, diode, gate driver) was designed by the team; the CC/CV control-loop firmware that decides what duty cycle to command, and the independent hard-limit protection that overrides it regardless of what the algorithm requests, were implemented.

6.2.1.2 GaN-Based Boost Converter with CC/CV Battery Charging Evaluation Criteria

Criterion	Why it matters
Control loop stability	Must reach and hold CC/CV targets without destructive oscillation or overshoot
Fault coverage independent of control mode	A bug in the control algorithm must not be the only thing standing between the circuit and an overvoltage/overcurrent event
Component selection for the power stage	Switching device, inductor, and diode must support ~100 kHz operation at the target power level
Code maintainability	Control logic must be testable independently of the PWM-generation mechanism underneath it

Table 7: Evaluation Criteria

6.2.1.3 GaN-Based Boost Converter with CC/CV Battery Charging Design Alternatives

1. Control algorithm: a full PID controller vs. a simple incremental step controller (step duty cycle up/down by a fixed amount each control tick, based on whether current/voltage are above or below target).
2. Switching device: GaN HEMT (Renesas TP65H150G4PS) vs. a conventional silicon MOSFET.
3. Boost diode: Schottky (1N5822) vs. a standard silicon rectifier.

6.2.1.4 GaN-Based Boost Converter with CC/CV Battery Charging Design Comparative Analysis and Conclusion

Control algorithm: The step controller was chosen over a full PID for this first working version. It has no proportional or derivative term, so it responds slowly and predictably rather than reacting hard to sudden sensor noise- a deliberate choice for safety while testing unvalidated hardware. It is also easy to reason about and tune, with one step-size constant. The architecture separates "decide what duty cycle" (ChargeController) from "safely apply whatever duty cycle it's told" (BoostConverter) - see Feature 3 for the PWM-generation mechanism itself, which is where the project's most significant technical problem (duty-cycle resolution) actually lived, not in this control algorithm.

Switching device: GaN was selected over silicon for its lower switching losses and higher achievable switching frequency at this power level, consistent with the project's wide-bandgap-semiconductor research goals (this project serves as a testbed for Texas State's UWBG Power Converter research initiative).

Boost diode: A Schottky diode (1N5822, 3 A) was chosen for its low forward-voltage drop and fast recovery time at the ~100 kHz switching frequency, where a standard silicon diode's slower reverse-recovery behavior would introduce unacceptable switching losses.

6.2.2 GaN-Based Boost Converter with CC/CV Battery Charging Block Diagram

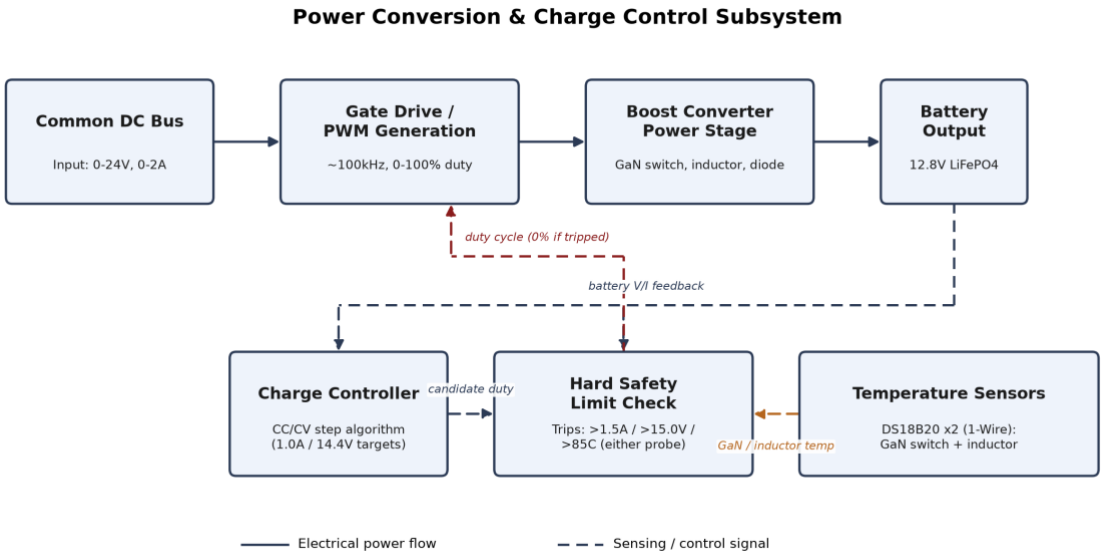


Figure 3: GaN-Based Boost Converter with CC/CV Battery Charging

As shown in Figure 3, this subsystem takes the common DC bus voltage/current as its electrical input and the battery's measured voltage/current as its control feedback, producing a PWM gate-drive signal as its only electrical output. The control chain mirrors the firmware's own call order: the Charge Controller computes a candidate duty cycle from the CC/CV algorithm, and the Hard Safety Limit Check (Feature 3) has the final say -it can override that candidate to 0% if voltage, current, or either temperature probe exceed the absolute limits, regardless of what the Charge Controller requested.

6.2.3 GaN-Based Boost Converter with CC/CV Battery Charging Interface

Inputs

Input	Source	Min	Max	Description
Common DC bus power	Solar + wind branches, combined after isolation diodes	0 V / 0 A	~24 V / 2 A (2S4P expansion case)	Variable renewable DC input to the boost converter
Battery feedback (voltage, current)	Battery-side INA226 sensor, via I2C	0 V / 0 A	36 V / 2 A (sensor configured range)	Closed-loop feedback used by the CC/CV algorithm

Table 8:Inputs

Outputs

Output	Destination	Min	Max	Description
Candidate duty cycle	Hard Safety Limit Check (Feature 3)	0%	80% (CC/CV ceiling, independent of the hardware's own 0-100% clamp)	The CC/CV algorithm's requested duty cycle, before the independent safety override

Regulated charge output	12.8 V LiFePO4 battery	0 V / 0 A	15.0 V / 1.5 A (hard safety limit); target 14.4 V / 1.0 A	CC/CV-regulated charging current/voltage

Table 9: outputs

6.3 Real-Time Monitoring and Display Subsystem

The Real-Time Monitoring and Display Subsystem generates the PWM signals driving both converters' gate drivers, enforces safety limits independently of any control algorithm, reads all system sensors, and presents live status on a touchscreen with touch-driven relay control.

6.3.1 Real-Time Monitoring and Display Design Analysis

The design analysis evaluates PWM generation method, microcontroller board, sensor and display libraries, screen layout, and relay control model based on switching performance, fault coverage, measurement accuracy, I2C capacity, and display usability.

6.3.1.1 Real-Time Monitoring and Display Objective

This is the subsystem that actually generates the PWM signals driving both the boost and buck converters' gate drivers, enforces safety limits independently of whatever any control algorithm requests, reads all system sensors (power and temperature), and presents that information on a touchscreen display -matching the SOW's listed responsibility: "INA226 power monitoring, PWM control, protection, and Arduino/software control development." It also provides touch-driven control of six relay-switched circuits (solar array, wind subsystem, and four demonstration loads), with an automatic power-availability check that can hold a load relay off even when the operator has requested it on.

6.3.1.2 Real-Time Monitoring and Display Evaluation Criteria

Criterion	Why it matters
Achievable PWM switching frequency (~100 kHz) and duty-cycle resolution	Matches both converters' component selection; determines how finely any control loop built on top can hold its targets
Fault coverage independent of control mode	A bug in either converter's control algorithm must not be the only thing standing between the circuit and an overvoltage/overcurrent/overtemperature event
Measurement range and accuracy	Must cover the project's 0-2 A operating range with enough resolution to be useful
I2C address availability	Up to 5 devices (4 active INA226 sensors + touch controller) share one bus without address collisions
Display readability and touch reliability	Values must be legible at demo distance; tap targets must be reliably distinguishable from accidental touches
Separation from control-algorithm logic	PWM generation, protection, and display code should each be testable independent of whatever control algorithm is driving them

Table 10: Evaluation Criteria

6.3.1.3 Real-Time Monitoring and Display Design Alternatives

1. PWM generation method: the microcontroller's built-in analogWrite() vs. direct configuration of a hardware timer's registers.
2. Target microcontroller board: Arduino Mega 2560 (per the SOW) vs. Arduino GIGA R1 WiFi.
3. Sensor library: a general-purpose INA226 Arduino library (e.g., Adafruit's) vs. a library exposing explicit shunt-resistor and current-range calibration (INA226_WE).
4. Display driver library: the display manufacturer's bundled demo library vs. a community-maintained Adafruit-compatible driver.
5. Screen layout: a single screen showing all values at once vs. a tabbed interface showing one category at a time.

6. Relay control model: fully automatic (sensor-driven) switching vs. touch-initiated switching with an automatic power-availability override.

6.3.1.4 Real-Time Monitoring and Display Design Comparative Analysis and Conclusion

PWM generation method: analogWrite() tops out around 490-980 Hz, far short of ~100 kHz, so direct timer configuration was used. AVR gives 160 duty steps; the Giga's TIM8 gives 2,400. Mbed's PWM API measured only ~10% steps on the scope, hence the switch to direct register control.
Microcontroller board: the Mega 2560 (SOW's original target) had unresolvable upload failures, so the project moved to the Giga R1 WiFi (STM32H7), requiring the PWM code to be ported. 100 kHz and ~0.042% resolution are confirmed on the Giga. Only the PWM module needed changing, validating the modular design.
Sensor library: INA226_WE was chosen for its explicit shunt and current-range calibration, needed for accurate readings at the project's 0.010 Ω shunt and 0-2 A range.
Display driver library: the bundled LCDWIKI demo library was rejected for outdated, compile-breaking code. Adafruit_ST7796S_kbv was used instead, confirmed working on a Mega2560 with this panel.
Screen layout: a tabbed interface (8 tabs) was chosen over one dense screen for legible readings and simple touch targets.
Relay control model: touch-initiated switching was chosen over full automation, except the 4 demo loads, which get an automatic power-availability veto, the same layered pattern as the hard safety limits. This keeps a human in the loop while still demonstrating automatic load-shedding.

6.3.2 Real-Time Monitoring and Display Block Diagram

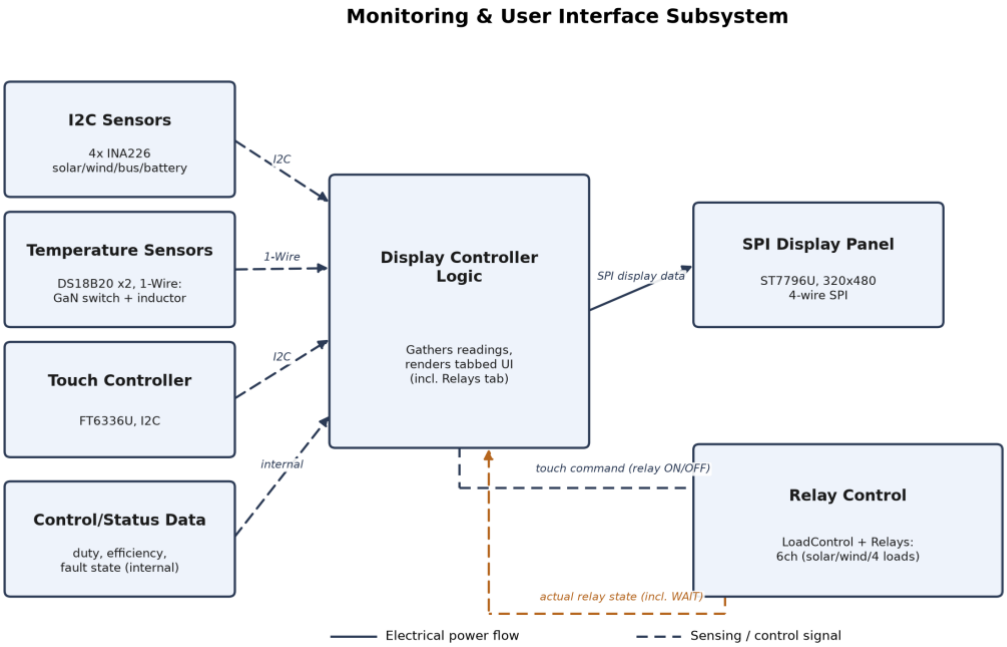


Figure 4: Real-Time Monitoring and Display -Block Diagram

As shown in Figure 4, this subsystem's inputs are the sensor readings and touch input, not the physical sensors themselves. The Control/Status Data input represents the candidate duty cycle and fault state fed in from Feature 2's CC/CV algorithm, which this subsystem's own Hard Safety Limit Check can override. The Relay Control block is bidirectional: touch commands flow out to it, and its

actual relay state (including a third "WAIT" state when a load is requested on but held off by the power-availability check) flows back for display.

6.3.3 Real-Time Monitoring and Display Interface

Inputs

Input	Source	Min	Max	Description
I2C sensor data	4 active INA226 modules + 1 reserved placeholder address	0 V / 0 A	36 V / 2 A (configured range)	Voltage/current/power, polled over the shared I2C bus
Temperature data	2x DS18B20 probes, 1-Wire	Ambient	85°C trip point	GaN switch and boost inductor temperatures
Touch input	FT6336U capacitive touch controller, I2C + dedicated lines	-	-	Single-touch coordinates, used for tab selection and relay button taps
Candidate duty cycle	Feature 2's CC/CV algorithm	0%	80%	Subject to override by this subsystem's hard safety limit check

Table 11:inputs

Outputs

Output	Destination	Min	Max	Description
Gate-drive PWM signals	TC4427 gate drivers (boost and buck)	0% duty	100% duty (clamped)	~100 kHz PWM; the only mechanism by which either converter's switching device is actually driven
Display data (4-wire SPI)	ST7796U display controller	-	-	Renders the tabbed UI: readings, duty/efficiency, fault status, relay button states
Relay switching commands	6-channel relay module (solar, wind, 4 demo loads)	0 (de-energized)	1 (energized)	Operator intent, gated by the automatic power-availability check for the 4 load relays

Table 12:Outputs

6.4 GaN-Based Buck Converter and 5 V Power Regulation Subsystem

The GaN-Based Buck Converter and 5 V Power Regulation Subsystem steps the battery's 12.8 V output down to a regulated 5 V bus for the demonstration loads, using a second GaN switch operating independently of the boost converter.

6.4.1 GaN-Based Buck Converter and 5 V Power Regulation Design Analysis

The design analysis evaluates topology validation, rectifying diode selection, and gate-drive timer selection based on output regulation accuracy, current-ceiling ratings, and independence from the boost converter's PWM.

6.4.1.1 GaN-Based Buck Converter and 5 V Power Regulation Objective

This subsystem steps the 12.8 V battery output down to a regulated 5 V DC bus supplying the model loads (data center, EV charging, LED lamp, and other demonstration loads). Unlike Feature 2's CC/CV battery charging, this is primarily constant-voltage regulation with a current ceiling: the converter's normal job is holding 5.0 V regardless of load, with an automatic override that caps total current if connected loads try to draw more than the design allows.

6.4.1.2 GaN-Based Buck Converter and 5 V Power Regulation Evaluation Criteria

Criterion	Why it matters
Output voltage regulation	Must hold 5.0 V across the expected load range without a separate control loop per load
Current ceiling matched to real component ratings	The ceiling must reflect what the actual power-stage components (diode, inductor) can safely deliver, not an assumed figure
Gate-drive timer independence from the boost converter	Driving a second GaN switch must not risk interfering with Feature 2's already-validated PWM generation

Table 13: Evaluation Criteria

6.4.1.3 GaN-Based Buck Converter and 5 V Power Regulation Design Alternatives

1. Topology validation method: hand-calculated component estimates vs. LTspice simulation before committing to real parts.
2. Rectifying diode: the originally simulated 1N5819 (1 A) vs. a higher-current Schottky already used elsewhere in the system (SB540, 5 A).
3. Gate-drive timer: sharing the boost converter's TIM8 peripheral (different channel) vs. a fully separate timer (TIM1).

6.4.1.4 GaN-Based Buck Converter and 5 V Power Regulation Design Comparative Analysis and Conclusion

Topology validation: the buck converter's design was validated in LTspice before ordering components, rather than relying on hand-calculated estimates. The simulated circuit (12 V input, EPC2204 GaN switch, 47 µH inductor, 220 µF output cap, 1N5819 diode, 100 kHz at 41.7% duty) settled to ~4.7-4.8 V at ~0.95 A into a 5 Ω load, consistent with the ideal buck relationship and confirming the topology before committing to hardware.
Rectifying diode: the simulated 1N5819 is rated for only ~1 A, but the real system's calculated load (EV charger + data center + LED lamp + other) totals ~1.6 A, rounded to a 2.0 A ceiling for margin (the same ~50% headroom pattern as Feature 2's current limit). The SB540 (5 A/40 V) was selected instead, both for margin above the ceiling and because it's already used elsewhere in the project for the solar/wind isolation diodes.
Gate-drive timer: a separate timer (STM32's TIM1) was selected instead of sharing Feature 2's TIM8, keeping this subsystem's PWM configuration fully independent and avoiding any risk of interfering with Feature 2's already-validated TIM8 configuration. The exact GPIO pin/alternate-function mapping for TIM1 Channel 2 hasn't been confirmed against the Giga's official pinout yet; it's pending bench verification via a standalone test sketch, the same verification discipline that caught Feature 3's PWM-resolution problem rather than trusting documentation alone.

6.4.2 GaN-Based Buck Converter and 5 V Power Regulation Block Diagram

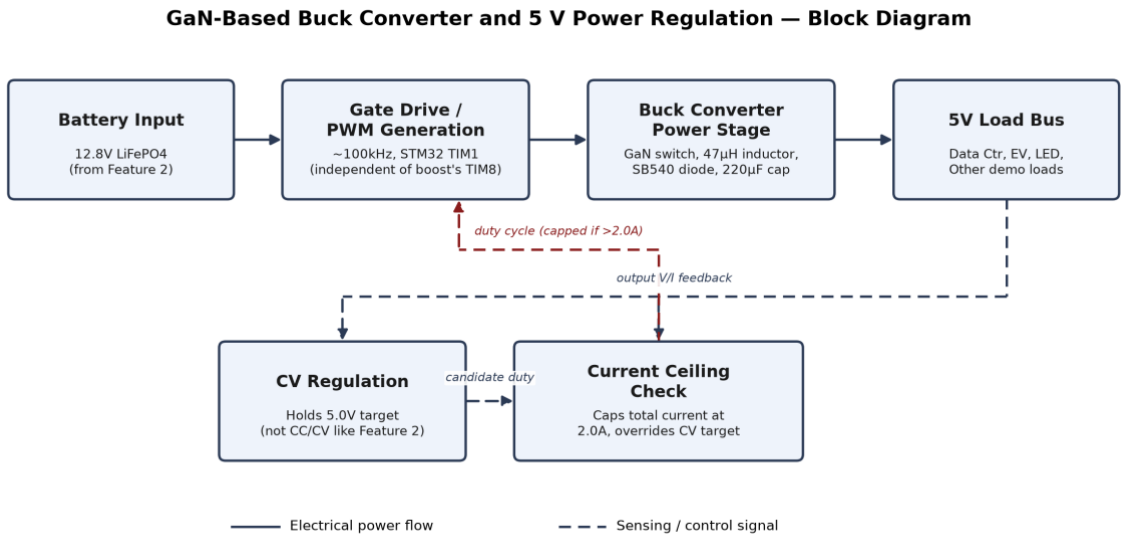


Figure 5:GaN-Based Buck Converter and 5 V Power Regulation

As shown in Figure 5, this subsystem's electrical input is the 12.8V battery output (Feature 2), and its only electrical output is the regulated 5V load bus. Unlike Feature 2's CC/CV charging control, the control chain here is CV-with-a-ceiling: CV Regulation computes a candidate duty cycle to hold 5.0V, and the Current Ceiling Check has the final say - it caps total output current at 2.0A (matched to the SB540 diode's real rating) regardless of what CV Regulation requested, the same layered-override pattern used in Feature 2's Hard Safety Limit Check.

6.4.3 GaN-Based Buck Converter and 5 V Power Regulation Interface

Inputs

Input	Source	Min	Max	Description
Battery output	12.8 V LiFePO4 battery (Feature 2's output)	0 V / 0 A	15.0 V / 1.5 A	The buck converter's input rail
Output voltage feedback	Load-bus INA226 sensor	0 V	36 V (sensor configured range)	Closed-loop feedback for the CV regulation target

Table 14: Inputs

Outputs

Output	Destination	Min	Max	Description
Gate-drive PWM signal	Dedicated TC4427 gate driver (buck stage)	0% duty	100% duty (clamped)	~100 kHz PWM, generated via STM32 TIM1 (pending pin verification)
Regulated 5 V load bus	Model loads (data center, EV charging, LED lamp, other)	0 V / 0 A	5.0 V / 2.0 A (calculated ceiling)	CV-regulated, with automatic current-ceiling override- not a symmetric CC/CV profile like Feature 2's battery charging

Table 15: Outputs

6.5 GaN device fabrication Subsystem

The GaN Device Fabrication Subsystem is responsible for developing and fabricating a GaN-based transistor using the university's cleanroom facilities. The fabricated device will be electrically characterized and evaluated for potential integration into the project's boost and buck converter stages.

6.5.1 GaN device fabrication Design Analysis

The design analysis evaluates potential GaN device fabrication approaches based on electrical performance, fabrication complexity, available cleanroom equipment, cost, and compatibility with the power conversion system.

6.5.1.1 GaN device fabrication Objective

The objective of this subsystem is to fabricate a functional GaN-based transistor using semiconductor microfabrication processes, including photolithography, thin-film deposition, and metal contact formation. The fabricated device will be electrically characterized to determine its suitability for use as a switching device in the project's GaN-based boost and buck converters.

6.5.1.2 GaN device fabrication Evaluation Criteria

Table 16: Evaluation Criteria

Criterion	Why It Matters
Electrical performance	The fabricated device must demonstrate transistor operation and sufficient switching capability for the intended application.
Fabrication feasibility	The selected fabrication process must be compatible with available university cleanroom equipment and materials.
Device reliability	The fabricated transistor should demonstrate repeatable electrical characteristics without excessive leakage current or premature breakdown.
Fabrication complexity	The process should minimize unnecessary fabrication steps and reduce the risk of manufacturing defects.
Cost and time	Fabrication must be achievable within the project's budget and second-semester development schedule.
Converter compatibility	The fabricated device must be evaluated for compatibility with the gate driver, voltage, and current requirements of the existing converters.

6.5.1.3 GaN device fabrication Design Alternatives

1. Fabrication of a GaN High Electron Mobility Transistor (HEMT): Fabricate a GaN-based HEMT using a suitable semiconductor heterostructure, photolithography, metal deposition, and contact formation. This approach provides hands-on experience with GaN device fabrication and allows the device to be electrically characterized.

2. Fabrication of a GaN MOS-HEMT: Fabricate a GaN transistor incorporating an insulating gate dielectric between the gate electrode and semiconductor structure. This approach may reduce gate leakage but introduces additional deposition steps and interface-quality requirements.

3. Commercial GaN Transistor: Use a commercially manufactured GaN power transistor instead of fabricating a device. This alternative provides predictable electrical specifications and simplifies converter integration but does not fulfill the project's semiconductor fabrication objective.

6.5.1.4 GaN device fabrication Design Comparative Analysis and Conclusion

Table 17: Alternatives analysis

Criterion	GaN HEMT	GaN MOS-HEMT	Commercial GaN
Electrical performance	Dependent on fabrication quality	Dependent on fabrication quality	Manufacturer-specified

Fabrication feasibility	Requires suitable GaN material and cleanroom processes	Requires additional dielectric processing	No fabrication required
Fabrication complexity	Moderate to high	High	Low
Cost and time	Dependent on wafer and equipment availability	Additional processing requirements	Lower integration risk
Educational fabrication objective	Fully supports	Fully supports	Does not satisfy
Converter compatibility	Requires testing	Requires testing	Can be selected from datasheet specifications

Proposed selection: Fabrication of a GaN HEMT is the preferred approach, provided the necessary starting wafer and fabrication equipment are available. Compared with a MOS-HEMT, a conventional HEMT avoids the additional gate dielectric deposition step, potentially reducing process complexity.

Commercial GaN transistors will remain available for initial converter development and as a reference during electrical characterization.

The fabricated device will be evaluated through current-voltage measurements, leakage current testing, and other electrical characterization procedures supported by the available laboratory equipment. Integration into the converters will depend on whether its measured electrical characteristics satisfy the required operating conditions.

6.5.2 GaN device fabrication Block Diagram

The GaN device fabrication subsystem follows a semiconductor fabrication process beginning with a suitable GaN-based wafer and ending with electrical characterization of the fabricated transistor. The process includes wafer preparation, photolithography, device isolation, metal contact formation, and device testing. The fabricated transistor will be evaluated for potential integration into the boost and buck converter circuits.

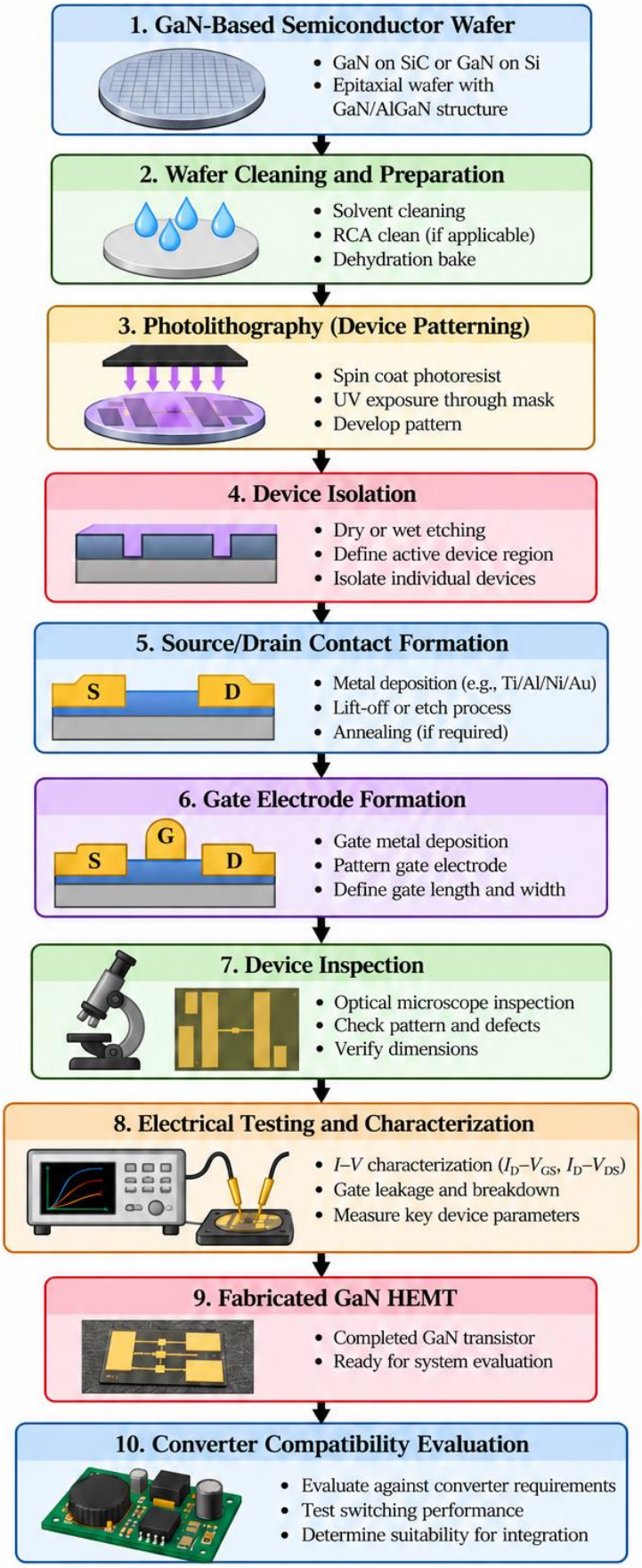


Figure 6: Block Digarm for GaN fabrication

6.6 GaN device fabrication Interface

Table 18: Inputs

Input	Source	Description
GaN semiconductor wafer	University laboratory	Starting semiconductor material used to fabricate the transistor.
Device design specifications	Boost and buck converter subsystems	Required voltage, current, and gate-drive characteristics used to evaluate device suitability.
Photolithography equipment	University cleanroom	Used to define device patterns and electrode regions.
Fabrication materials	University cleanroom	Photoresist, processing chemicals, and metal deposition materials required for fabrication.
Fabrication equipment	University cleanroom	Equipment used for approved etching, deposition, and contact formation processes.

Table 19: Outputs

Output	Destination	Description
Fabricated GaN transistor	Electrical characterization laboratory	Semiconductor device produced through the fabrication process.
Electrical characterization data	GaN device fabrication subsystem	Measured current-voltage characteristics, gate leakage, and other device parameters.
Device performance results	Boost and buck converter subsystems	Electrical test results used to determine whether the fabricated transistor meets converter requirements.
Qualified GaN switching device	Power conversion subsystem	Fabricated transistor considered for integration into the converter circuit if electrical testing confirms compatibility.

6.7 Safety, Security and Administrative Functions

Safety: Both GaN converters (boost, Feature 2; buck, Feature 4) can be commanded by either an automatic control algorithm or a manual bench-test mode. All modes share one independent hard-limit check, implemented in Feature 3: if battery voltage exceeds 15.0 V, current exceeds 1.5 A, or either temperature probe (GaN switch, boost inductor) exceeds 85°C, the boost converter's output is immediately forced to 0% duty cycle, regardless of which mode commanded the unsafe value. This exists specifically because manual test mode has no other protection: a person typing a duty-cycle value by hand has nothing else stopping an unsafe command from reaching the gate driver. The buck converter (Feature 4) carries an analogous current-ceiling override, capping output current at 2.0 A regardless of load demand.

Two further automatic-mode checks protect against conditions the control algorithm alone cannot detect: (1) if the battery sensor stops responding on I2C, the system cannot distinguish "no feedback" from "feedback reading zero," so it treats a missing sensor as an immediate fault; (2) if

duty cycle is pinned at its ceiling for more than 3 seconds without reaching the target charging current, this is treated as a likely open-circuit or disconnected-battery condition.

All faults latch: once tripped, the output stays forced to 0% and does not auto-retry, even if the triggering condition later goes away on its own. Clearing a fault requires a manual reset (power cycle or reset button). In manual test mode, any typed duty-cycle command is refused outright once a fault is latched, so a bench operator cannot accidentally override the safety trip.

Relay safety: the six relay-switched circuits (solar array, wind subsystem, and four demonstration loads) are deliberately scoped to exclude anything the charge controller needs to stay connected to function (e.g. battery-to-charger) or that is meant as a human bench-test disconnect (e.g. overall system master power, which cannot be relay-controlled by the Arduino it powers without creating a self-disconnect paradox). Relay channel assignment keeps the Giga's GPIO pins only ever driving the relay module's opto-isolated logic input, never carrying relay-coil current directly.

Administrative functions: the choice between manual test mode and automatic charging mode is a compile-time setting (MANUAL_PWM_TEST_MODE in the firmware configuration), not a runtime option. Changing it requires re-uploading the firmware- a deliberate restriction functioning as an implicit access control over this debug capability.

Security: this subsystem does not store user accounts, credentials, or personally identifiable data. A read-only local-network status dashboard (hosted by the Giga's own WiFi hardware) is planned but not yet implemented in firmware; relay control is deliberately excluded from that dashboard's scope and remains touchscreen-only, so no network-originated command can switch power on this system. The touchscreen itself has no administrative login; all sensor readings and fault status are read-only to the user, and the only "write" actions available at the display are tab navigation and the six relay buttons, bounded by the safety scoping described above.

7 Conclusion

This project aims to develop a hybrid solar-wind power conversion and energy storage system using GaN-based boost and buck converters. The design combines solar and wind energy sources, CC/CV battery charging, a 12.8 V LiFePO4 battery, regulated 5 V power distribution, and real-time monitoring in a physical smart-city model.

The project will be evaluated through source characterization, converter testing, battery charging verification, monitoring checks, and full-system integration. The test procedures in Appendix A will provide evidence of the system's performance against the design objectives established in the Introduction. Key evaluation areas include renewable-source output, charging voltage and current regulation, 5 V output stability, and reliable operation of the integrated subsystems.

Because the project is still under development, final conclusions regarding performance and successful completion of the design objectives will be made after testing. If the system does not meet its intended operating targets, further work may include adjusting converter components or control parameters, improving source configuration, and addressing issues identified during integration. The final assessment will be based on measured results and documented limitations rather than assumed performance.

8 Project Alignment Matrix

Table 20:Knowledge Alignment Matrix

Course No.	**Core knowledge**	**Specific knowledge incorporated by team**
EE 3350 (Microelectronics)	*Design and analysis of active devices and equivalent circuits*	The team applies semiconductor switching principles when selecting and using GaN transistors, diodes, and gate-driver circuits in the boost and buck converters.
EE 3370 (Signals and Systems)	*Frequency domain representation of signals and frequency response, transfer functions*	The team applies time-varying signal concepts when working with PWM switching signals and evaluating converter behavior.
EE 3320 (Microprocessors)	*Principles of operation and applications of microprocessors*	The team uses a microcontroller to read voltage and current sensors, implement charging control, and display system measurements.
EE 4352 (Introduction to VLSI Design)	*Analysis and design of CMOS integrated circuits*	The project has limited direct application of CMOS IC design. General knowledge of integrated circuits supports understanding electronic

		components, but the project does not involve designing a custom CMOS IC.
EE 4356 (Power Electronics)	*Circuits and devices for the control and conversion of electric power*	The team designs and tests DC-DC boost and buck converters, uses PWM switching, and evaluates voltage regulation and power conversion.
EE 4357 (Intro to Power Systems)	*This course introduces the analysis of various elements of power systems*	The team applies basic power-flow and distribution concepts when combining solar and wind DC sources, storing energy in a battery, and distributing regulated power to model-city loads.
EE 4360 (Linear Control Systems)	*Performance and stability of feedback-based control systems*	The team applies feedback-control concepts to the CC/CV battery charging system to regulate charging current and voltage.
EE 4370 (Communications Systems)	*Transmission of signals through linear systems, analog, and digital modulation, and noise*	This course has limited direct application because the project does not use a dedicated communications link or modulation system. General signal-integrity concepts may be relevant to sensor and control signals.

Table 21: Design Considerations Matrix

Design Considerations	Project Specific Implementation
Public health, safety, and welfare	The design uses low-voltage DC power, fuses, appropriate wiring, and battery charging limits. Testing will check for short circuits, excessive current, and overheating.
Global	Components are sourced from available suppliers, and delivery times and availability affect project scheduling. The prototype demonstrates renewable energy concepts but is not designed for direct grid connection.
Cultural	The touchscreen will use clear labels and familiar electrical units. No culture-specific interface features are required for this demonstration.
Social	The project demonstrates renewable energy conversion and storage, supporting student learning about sustainable electrical systems.
Environmental	Solar and wind energy are used to demonstrate renewable generation. Efficient power conversion and reusable, replaceable components are considered where practical.

Economic	The team uses commercially available components and a small-scale prototype to keep costs manageable while demonstrating the main functions of the system.

Table 22: Ethical Considerations Matrix

Ethical Considerations	Project Specific Implementation
Global	Solar and wind energy systems may be useful in different regions, but their practical performance depends on local sunlight, wind availability, cost, and maintenance. This prototype is an educational demonstration, not a field-ready energy system.
Economic	The team considers component cost, availability, and performance when selecting parts. The prototype's results will help evaluate the trade-offs involved in a larger implementation.
Environmental	The project demonstrates renewable energy and considers power-conversion losses. The team should handle batteries and electronic components responsibly and avoid unnecessary waste during prototyping.
Societal	The project can improve understanding of renewable energy and energy storage. Electrical safety is considered through protective components and testing; no personal data collection or remote monitoring is part of the stated design.

9 References

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[7] Texas State University, *Electrical Engineering (EE) Course Catalog*. Course descriptions for EE 3350, EE 3370, EE 3320, EE 4356, EE 4357, EE 4360, and EE 4370.

[8] E1.07 Power Rangers, *Project Materials Lists*, SD MATERIALS list 1 and SD MATERIALS LIST 2. Internal project documents, 2026.

10 Approvals

The signatures of the people below indicate an understanding in the purpose and content of this document by those signing it. By signing this document, you indicate that you approve of the Project Plan.

Approver Name	Title	Signature	Date
Avash Aryal	Project Manager		
Alok Parajuli	Team Member		
Ashok Paudel	Team Member		
Michelle Su	Team Member		
David Bendeck Cortes	D2 Project Manager		
Dr. Dinusha Herath Mudiyanselage	Faculty Advisor		
Dr. Dinusha Herath Mudiyanselage	Sponsor		
Dr. Mark Welker	Instructor		

10.1 Contingencies

This section contains any approval contingencies as noted by the Product Sponsor, Advisor, Mentor, or Instructor.

Approver Name	Contingency

Section	Author	Word Count
Abstract	Alok Parajuli	
Introduction	Alok Parajuli	
System Architecture	Avash Aryal and Michelle Su	
Solar and Wind Energy Harvesting Subsystem	Alok Parajuli	

GaN-Based Boost Converter with CC/CV Battery Charging	Avash Aryal and Ashok Paudel	
Real-Time Monitoring and Display	Ashok Paudel	
GaN-Based Buck Converter and 5 V Power Regulation	Avash Aryal and Michelle Su	
Safety, Security and Administrative Functions	Avash Aryal and Michelle Su	

11 Appendix A: Test Procedures

Validation testing will be conducted at both the subsystem and full-system levels to verify that each feature meets its functional requirements before final integration. Individual subsystem tests will evaluate solar and wind power generation, boost converter operation and CC/CV battery charging, real-time monitoring and control, and regulated 5 V power delivery. Testing will begin with individual bench tests using measured electrical outputs and representative loads, followed by integrated testing of the complete system. Full-system tests will verify that renewable power sources supply the common DC bus, the battery stores energy, the buck converter maintains the required 5 V output, and the monitoring system displays measurements and controls the demonstration loads. Each test will have a designated directly responsible individual (DRI), with results documented to identify performance limitations and guide necessary adjustments before final system validation.

Test Plan Overview	
Solar and Wind Energy Harvesting Subsystem Unit Tests	**DRI**
1. Solar Panel Output Characterization: Measure the solar array's output voltage and current under available lighting conditions to verify that it produces usable DC power.	Alok Parajuli
2. Wind Generator Output Test: Operate the external drive motor and belt-and-pulley assembly, then measure the combined generator output voltage and current.	Alok Parajuli
Real-Time Monitoring and Display Subsystem Unit Tests	**DRI**
1. Sensor Measurement and Display Test: Compare the INA226 voltage and current readings against reference measurements and verify that the touchscreen displays the system measurements correctly.	Ashok Paudel
2. PWM Generation and Protection Test: Verify approximately 100 kHz PWM operation, confirm that duty-cycle adjustments function correctly, and test that the safety limit can override the control command.	Ashok Paudel
GaN-Based Buck Converter and 5 V Power Regulation Subsystem Unit Tests	**DRI**
1. 5 V Output Regulation Test: Supply the buck converter from the 12.8 V battery and measure its output with different loads to verify that the output remains near 5.0 V.	Michelle Su
2. Output Current Limit Test: Increase the connected load demand while monitoring output current to verify that the converter's current-limiting behaviour protects the power stage when the load exceeds its allowable operating range.	Michelle Su
GaN-Based Boost Converter and CC/CV Battery Charging Subsystem Unit Tests	**DRI**
1. CC/CV Battery Charging Test: Verify that the boost converter maintains approximately 1.0 A constant-current charging until the	Avash Aryal

battery approaches 14.4 V, then transitions to constant-voltage regulation.	
2. Converter Safety Limit Test: Verify that the independent safety mechanism overrides the requested PWM duty cycle when the configured voltage, current, or temperature limit is exceeded.	Avash Aryal
Overall System Integration Tests	**DRI**
1. Renewable Energy to Battery Charging Test: Operate the solar array and wind generator, verify their outputs reach the common DC bus, and confirm that the boost converter charges the battery according to the CC/CV profile.	Avash Aryal & Alok Parajuli
2. Battery-to-Load Power Delivery Test: Operate the buck converter from the battery and power the intended 5 V demonstration loads while verifying output voltage, current, and system monitoring readings.	Michelle Su
Integrated Monitoring and Load Control Test: Use the touchscreen to request operation of the demonstration loads and verify that the monitoring system displays the corresponding measurements and prevents load activation when insufficient power is available.	Ashok Paudel

Table 23

Feature 3 (Real-Time Monitoring and Display) Subsystem Unit Tests

#	Test	DRI
1	100 kHz PWM Frequency and Duty-Cycle Verification	Ashok Paudel
2	Hard Safety Limit Trip Test	Ashok Paudel
3	INA226 Sensor Accuracy Verification	Ashok Paudel
4	Touch Tab Navigation and Relay Control Test	Ashok Paudel

Feature 4 (GaN-Based Buck Converter and 5 V Power Regulation) Subsystem Unit Tests

#	Test	DRI
1	Buck Converter Output Voltage and Current-Ceiling Verification	Ashok Paudel and Avash Aryal

Overall System Integration Tests (minimum 3 required by the template -- drafted candidates, not yet confirmed with the full team)

#	Test	DRI
1	Combined solar + wind input successfully boosts and charges the battery, confirmed live on the touchscreen display	(needs team confirmation -- spans Feature 1, 2, 3)
2	Battery powers the 5 V load bus via the buck converter when renewable sources are inactive, confirmed on the display	(needs team confirmation -- spans Feature 2, 4, 5)
3	An overtemperature or overcurrent fault on the boost converter immediately disables output and is visible on the Status tab within the same second	(needs team confirmation -- spans Feature 2, 3)

11.1 Solar and Wind Energy Harvesting Subsystem Unit Tests

11.1.1 Solar Panel Output Characterization

Test 11.1.1		Written by: Michelle Su 10/19/2026
Subsystem: Solar and Wind Energy Harvesting - Solar Panel Output Characterization		Tested by: TBA
Equipment Required:	*Multimeter, solar panel array (four 6 V, 3 W photovoltaic panels), appropriate resistive load, connecting wires, and prototype system.*	
Description: *Confirms that the solar panel array produces usable DC power under available lighting conditions. The output voltage and current are measured under open-circuit and loaded conditions to determine whether the array can provide power to the downstream conversion and battery charging system.*		
Overall Results: TBA		

#	Procedure	Expected Result	Actual Result	Comments:
1	*Expose the solar panel array to available lighting conditions and measure its open-circuit output voltage using a multimeter.*	*12 V*	*TBA*	*TBA*
2	*Connect an appropriate load to the solar panel array and measure the output voltage and current.*	*1 A*	*TBA*	*TBA*
3	*Monitor the solar panel output under consistent lighting conditions and record voltage and current to measure the power output.*	*12 W*	*TBA*	*TBA*

Overall result = *TBA*
This test procedure could not be completed during the initial evaluation period due to equipment constraints and active hardware revisions. The test remains unverified at this time, and full validation of these performance specifications will be conducted during the next scheduled round of subsystem testing.

11.1.2 Wind Generator Output Test

Test *11.1.2*		Written by: Michelle *Su 10/19/2026*
Subsystem: Solar and Wind Energy Harvesting		Tested by: Alok Parajuli
Equipment Required:	*Multimeter, external drive motor, adjustable low-voltage DC PWM motor speed controller, three windmill assemblies with six R280 DC motors (three lower drive motors and three blade-mounted generator motors), three drive discs, rubber bands, appropriate resistive load, connecting wires, and prototype system with isolation diodes.*	
Description: Confirms that the three windmill assemblies produce usable electrical power when mechanically driven. The external drive motor powers the mechanical drive system, while an adjustable DC PWM motor speed controller regulates the driving speed to prevent the windmill blades from rotating too fast. Each lower motor uses a drive disc and rubber band to transfer rotation to the blade-mounted generator motor. The voltage and current produced by the three generators are measured to evaluate their combined output and compatibility with the common DC bus.		

Overall Results: TBA

#	Procedure	Expected Result	Actual Result	Comments:
1	*Operate the external drive motor using the adjustable PWM speed controller. Adjust the speed until all three windmill blades rotate consistently. Measure the voltage produced by each generator and the combined output voltage.*	*All three windmills rotate at a controlled speed, and each generator produces measurable DC voltage.*	*TBA*	*TBA*
2	*Connect an appropriate resistive load to the combined generator output. Measure the output voltage and current at the selected motor speed, and calculate the total output power.*	*The generator assembly supplies measurable electrical power under load. Voltage, current, and calculated power are recorded.*	*TBA*	*TBA*
3	*Run the assembly continuously at the selected PWM speed setting. Monitor generator output and inspect the drive discs, rubber bands, and motor connections for slippage or inconsistent rotation.*	*All three windmills operate consistently without significant slippage or unexpected interruptions. Electrical output remains measurable throughout the test.*	*TBA*	*TBA*

Overall result = TBA

This test procedure could not be completed during the initial evaluation period due to equipment constraints and active hardware revisions. The test remains unverified at this time, and full validation of these performance specifications will be conducted during the next scheduled round of subsystem testing.

11.2 GaN-Based Boost Converter and CC/CV Battery Charging Subsystem Unit Tests

11.2.1 GaN-Based Boost Converter and CC/CV Battery Charging

Test 11.2.1		Written by: Michelle Su 10/19/2026
Subsystem: Boost Converter		Tested by: TBA
Equipment Required:	*Oscilloscope with 10× probes and a 20 MHz bandwidth limit, multimeter, resistive load (25 Ω and approximately 13.64 Ω), adjustable DC power supply, IR2104 gate driver, TP65H150G4PS GaN transistor, driver power supply (approximately 12 V), and PWM control source.*	
Description: *Confirms that the GaN-based boost converter increases the input voltage to a higher DC output voltage under different input and load conditions. The converter was tested at 100 kHz with different PWM duty cycles, and output voltage, current, power, efficiency, and switching ripple were recorded. The results demonstrate boost conversion and approximately 1 A output capability under the tested heavy-load condition. The uploaded results do not document battery charging tests, so constant-current (CC) and constant-voltage (CV) charging operation cannot yet be confirmed.*		
Overall Results: PASS		

#	Procedure	Expected Result	Actual Result	Comments:
1	*Apply a 5 V input to the boost converter with a 25 Ω load. Test at 10% and 20% PWM duty cycle and measure output voltage and current.*	*Output voltage is higher than the input voltage, demonstrating boost conversion.*	*10%: 5.23 V, 0.209 A, 1.094 W* *20%: 5.87 V, 0.235 A, 1.378 W*	*Passed. Boost conversion was demonstrated at both duty-cycle settings, with recorded efficiencies of 93.8% and 92.9%, respectively.*
2	*Apply a 10 V input with a 25 Ω load. Test at approximately 20% and 30% PWM duty cycle and record the output voltage and current.*	*Output voltage increases above the input voltage and responds to changes in PWM duty cycle.*	*20%: 11.91 V, 0.476 A, 5.674 W* *30%: 13.67 V, 0.547 A, 7.475 W*	*Passed. Output increased from 11.91 V to 13.67 V as the duty cycle increased. The 30% test was repeated at 13.69 V and 0.548 A, confirming repeatability.*
3	*Increase the load using the approximately 13.64 Ω load and measure the output voltage, current, and power at different duty-cycle settings.*	*Converter maintains boost operation and supplies increased output current under heavier loads.*	*Approximately 20%: 11.53 V, 0.845 A* *25%: 12.73 V, 0.933 A* *30%: 13.65 V, 1.001 A*	*Passed. The converter supplied approximately 1.0 A at the highest recorded output point, delivering 13.66 W with a reported main-stage efficiency of 94.9%.*
4	*Measure the output waveform using an oscilloscope with a 10× probe and 20 MHz bandwidth limit. Record peak-to-peak ripple and RMS measurements.*	*Output waveform remains sufficiently stable, with switching ripple within the*	*At 5 V input: 1.10- 1.15 V peak-to-peak, RMS 35.6- 46.8 mV.*	*Failed pending ripple acceptance criteria. Switching spikes and ringing were observed, and peak-to-peak measurements were dominated by switching*

		design's acceptable limits.	*At 10 V input: approximately 1.74 V peak-to-peak, RMS 77 mV.* *Heavy load: approximately 1.88 V peak-to-peak, RMS greater than 93 mV.*	*transients. Further probing and layout improvements are needed before confirming ripple compliance. As shortened ground measurement probe reduced the ripple significantly for buck converter, we are hoping the same for boost converter.*
5	*Connect a battery under approved test conditions and measure charging current while the charger operates in constant-current (CC) mode.*	*Charging current remains regulated at the configured current limit.*	TBA	*TBA*
6	*Monitor battery voltage and charging current as the charger approaches the configured voltage setpoint and enters constant-voltage (CV) mode.*	*Charger maintains the specified battery voltage while charging current decreases.*	*TBA*	*TBA*

Overall result = Partial *Pass*

The boost converter successfully increased the input voltage and delivered approximately 13.65 V at 1.001 A, demonstrating an output power of approximately 13.66 W. Repeated tests at a 10 V input confirmed consistent output under the tested conditions. However, oscilloscope measurements showed switching spikes and ringing that require further investigation to determine whether the output ripple meets the design requirements. The CC/CV battery charging subsystem was not tested as part of these measurements. Additional testing will be performed to verify battery charging current regulation, voltage regulation, and the transition between constant-current and constant-voltage modes.

11.3 Real-Time Monitoring and Display Subsystem Unit Tests

11.3.1 100 kHz PWM Frequency and Duty-Cycle Verification

Test 11.3.1		Written by: Ashok Paudel 10/19/2026
Subsystem: Real-Time Monitoring and Display		Tested by: Ashok Paudel
Equipment Required:	*Oscilloscope with 10x probe, Arduino GIGA R1 WiFi with test sketch loaded, USB cable*	
Description: *Confirms the boost converter's gate-drive PWM output runs at the design target of 100 kHz, and that typed duty-cycle commands (0-100%) produce the expected, distinguishable output duty cycles.*		
Overall Results: PASS- Frequency and duty-cycle resolution both confirmed		

#	Procedure	Expected Result	Actual Result	Comments:
1	*Probe the gate-drive pin with the board running the manual PWM test sketch; measure period with the oscilloscope.*	*100 kHz (10 µs period)*	*100.01-100.02 kHz measured*	*Passed*
2	*Type 25, 26, 27, 28, 29, 30 into Serial Monitor; measure duty cycle at each.*	*Each value visibly distinct*	*5.02 V, 0.368 A, 1.848 W*	*Passed after the TIM8 rework; see §6.3.1.4*

Overall result = Pass

The gate-drive PWM output for the boost converter successfully met the 100 *kHz design target, measuring between* 100.01 *kHz and* 100.02 *kHz (corresponding to the expected* 10 *µs period). Sequential duty-cycle commands (*25% *to* 30%*) sent via the Serial Monitor produced clearly distinguishable PWM output duty cycles following the required TIM8 peripheral rework (detailed in §6.3.1.4). Under these test settings, the subsystem achieved output operating parameters of* 5.02 *V,* 0.368 *A, and* 1.848 *W, fully verifying both frequency precision and duty-cycle resolution.*

11.3.2 Hard Safety Limit Trip Test

Test 11.3.2		Written by: *Ashok Paudel*
Subsystem: Real-Time Monitoring and Display		Tested by: TBA
Equipment Required:	*Current-limited bench power supply, multimeter, battery-side INA226 sensor wired and powered*	
Description: *Confirms the boost converter's gate-drive PWM output runs at the design target of 100 kHz, and that typed duty-cycle commands (0-100%) produce the expected, distinguishable output duty cycles.*		
Overall Results: Could not be completed -not yet run on hardware		

#	Procedure	Expected Result	Actual Result	Comments:
1	With duty cycle at 0%, apply 15.5 V to the battery sensor's bus terminals from the bench supply	Fault message appears	*TBA*	*TBA*

		within ~50 ms; duty forced to 0%		
2	Lower supply voltage back below the limit without resetting the board	Fault remains latched (does not auto-clear)	*TBA*	*TBA*
3	Reset the board	Fault clears; normal operation resumes	*TBA*	*TBA*

Overall result = TBA
This test procedure could not be completed during the initial evaluation period due to equipment constraints and active hardware revisions. The test remains unverified at this time, and full validation of these performance specifications will be conducted during the next scheduled round of subsystem testing.

11.3.3 INA226 Sensor Accuracy Verification

Test 11.3.3		Written by: *Ashok Paudel*
Subsystem: Real-Time Monitoring and Display		Tested by: TBA
Equipment Required:	*Digital multimeter, bench power supply, known 1 kΩ precision load resistor*	
Description: *Confirms each INA226 sensor's reported voltage and current match independent multimeter measurements within tolerance, for a known applied voltage and load.*		
Overall Results: Could not be completed -not yet run on hardware		

#	Procedure	Expected Result	Actual Result	Comments:
1	Apply 12.00 V across a sensor's bus terminals from the bench supply; compare multimeter and sensor-reported voltage	Within ±1% or ±0.05 V	*TBA*	*TBA*
2	Apply 12.00 V across a sensor's bus terminals from the bench supply; compare multimeter and sensor-reported voltage	Within ±2% or ±5 mA	*TBA*	*TBA*

Overall result = TBA
This test procedure could not be completed during the initial evaluation period due to equipment constraints and active hardware revisions. The test remains unverified at this time, and full validation of these performance specifications will be conducted during the next scheduled round of subsystem testing.

11.3.4 Touch Tab Navigation and Relay Control Test

Test 11.3.4		Written by: *Ashok Paudel*
Subsystem: Real-Time Monitoring and Display		Tested by: TBA
Equipment Required:	*Assembled display/touch panel, Arduino GIGA R1 WiFi with firmware loaded, relay module wired*	
Description: *Confirms that tapping each tab switches to the corresponding category, taps below the tab bar have no effect except on the Relays tab, and relay buttons reflect ON/OFF/WAIT state correctly.*		
Overall Results: Could not be completed -display, touch controller, and relay module not yet verified on real hardware		

#	Procedure	Expected Result	Actual Result	Comments:
1	Tap each tab in turn	Active tab highlights; correct category's values display	*TBA*	*TBA*
2	Tap within the content area on a non-Relays tab	No change occurs	*TBA*	*TBA*
3	On the Relays tab, tap the Solar and Wind buttons	Each toggles directly between grey (OFF) and green (ON) -- no WAIT state possible	*TBA*	*TBA*
4	On the Relays tab, tap a load button while battery power is below the configured threshold	Button shows yellow "WAIT", relay does not energize	*TBA*	*TBA*

Overall result = TBA
This test procedure could not be completed during the initial evaluation period due to equipment constraints and active hardware revisions. The test remains unverified at this time, and full validation of these performance specifications will be conducted during the next scheduled round of subsystem testing.

11.4 GaN-Based Buck Converter and 5 V Power Regulation Subsystem Unit Tests

11.4.1 GaN-Based GaN-Based Buck Converter and 5 V Power Regulation

Test 11.4.1		Written by: Michelle Su 10/19/2026
Subsystem: Buck Converter		Tested by: Avash 10/16/2026
Equipment Required:	*Oscilloscope, electronic/resistive load bank (approximately 13.64 Ω), multimeter, Arduino GIGA R1 WiFi with buck converter firmware loaded, IR2104 high-side gate driver, approximately 12 V driver supply, and oscilloscope probes.*	
Description: *Confirms that the GaN-based buck converter steps down a 10 V input to a regulated 5 V output using PWM control. Output voltage, output current, power, and efficiency are measured at different PWM duty-cycle settings. The final test point verifies that the converter can produce approximately 5 V under the tested load.*		
Overall Results: PASS		

#	Procedure	Expected Result	Actual Result	Comments:
1	*Apply a 10 V input and a resistive load of approximately 13.64 Ω. Measure the output voltage at 20%, 30%, and 40% PWM duty cycle.*	*Output voltage increases as the PWM duty cycle increases.*	*20%: 1.44 V 30%: 2.26 V 40%: 3.20 V*	*Passed. Output voltage increased with PWM duty cycle, demonstrating buck converter operation.*
2	*Increase the PWM duty cycle to approximately 58% and measure the output voltage and current.*	*Output approaches the 5 V regulation target under the test load.*	*5.02 V, 0.368 A, 1.848 W*	*Passed. Output reached approximately 5 V at 58% PWM duty cycle.*
3	*Increase the PWM duty cycle to 60% and measure the output voltage and current. Compare the result with the 5 V regulation target.*	*Output remains within the specified 5 V regulation tolerance.*	*5.22 V, 0.383 A, 1.998 W*	*Passed. Output exceeded the 5 V target by only 0.22 V at 60% PWM.*
4	*Increase the load toward the intended 2.0 A output current and monitor output voltage and current.*	*Output current approaches the configured 2.0 A limit without exceeding it.*	*Not tested at 2.0 A; maximum recorded current was 0.383 A.*	*Could not be completed. The available measurements did not reach the intended 2.0 A current limit, so current limiting remains unverified.*
5	*Measure the output voltage ripple of the buck converter at steady state using an oscilloscope with the probe ground pin shortened.*	*Low output ripple voltage confirming stable*	*Pk-Pk: 48.00 mV, Freq: 100.31 kHz,*	*Passed. Shortening the ground pin minimized measurement pickup, yielding a clean peak-to-peak ripple of 48.00 mV at ~100.31 kHz.*

		switching operation.	*RMS: 11.51 mV,* *Mean: 3.22 mV.*	

Overall result = Partial Pass

The buck converter successfully demonstrated step-down operation, stepping down the 10 V input to approximately 5 V at a 58% PWM duty cycle (producing 5.02 V at 0.368 A). At 60% PWM, the output reached 5.22 V. Steady-state ripple testing with a shortened probe ground connection confirmed low output voltage ripple, measuring 48.00 mV peak-to-peak (11.51 mV RMS) at 100.31 kHz. However, testing at the intended 2.0 A current limit could not be completed, leaving current limiting unverified. Further testing under higher load conditions is required to verify full power and current limit specifications.